

From Sparse Selection to Physical Savings: A Unified Runtime and Measurement Contract for Progressive Retrieval Attention

PRA Research Program

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Abstract

Progressive Retrieval Attention (PRA) lets a query select a bounded subset of URI-addressed, model-native key/value memory. Earlier work established the model interface, external-memory lifecycle, typed capability records, and compact typed-result backing, but those parts did not yet form one product surface, and logical sparsity did not by itself establish lower latency or accelerator memory. We introduce a unified runtime that preserves the existing Hugging Face API while adding an engine-neutral four-axis execution policy, request-owned selection plans, versioned systems configuration, authenticated sessions, lazy tool and skill disclosure, independently authorized tool execution, session-scoped result compaction, size-adaptive native result indexing, lazy selected-region native encoding, durable task-aware agent state, selective replay, deduplicated K/V interval planning, physical byte accounting, a versioned cross-model product-profile registry, bounded caching, profiling, and a scheduler-unaware serving handoff. A standalone gateway now separates agent semantics, PRA mediation, and engine-native state through a JSON logical-reference contract and explicit capability negotiation and streams HF-backed responses with request-owned cleanup. The gateway now resolves engine-specific cache/session profiles, exercises prepare/close lifecycles, preserves append-only G10 prefixes, and transports message and resource deltas without conflating conventional prefix K/V with PRA-native state. A five-turn deterministic profile preserves the prior serialized prefix on all four eligible turns; physical cache hits remain unmeasured. A native-reference audit found and corrected a positional collision: external keys retained source-relative RoPE coordinates while direct queries restarted at zero. On a random tiny Qwen fixture, the corrected full-reference path matches visible-prefix logits to 5.97×10^{-8} maximum absolute error. On three one-token continuation cases, full native replay preserves visible-prefix top tokens and answer accuracy for Qwen3-0.6B and a public Llama-3.2-1B weight mirror. Gemma-3-1B preserves disabled behavior, native MQA shape, positions, and decode lifetime, but its four global PRA consumers do not reproduce the semantics of a visible prefix passing through 22 unchanged local sliding layers; we report that family as partial topology. The runtime now separates address encoding, detail-K/V encoding, request-time routing, and memory consumption. A corrected 102-row layer calibration finds that no sparse profile is portable across the three families: all eligible layers are the balanced correctness profile for each model, while Qwen’s last-24 profile is only a small-cohort quality-max candidate. DeepSeek Harness and Pi bridges preserve typed tool-result identity through explicit ordinary-engine fallback in all 10 contract cases. In a five-seed tiny-HF mechanism profile, token-shared selection required 2.8 routing operations per request, versus 8.4 for token-per-layer, a $3.0\times$ reduction with three active layers. On a CUDA 12.6 host with an NVIDIA GTX 950M, gathering 512 of 8,192 native-K/V tokens takes a median 0.253 ms at batch one, and packing eight selected intervals takes 0.415 ms. The same packed payload costs 0.424 ms when HBM-resident, 1.592 ms from pinned CPU memory, and 2.157 ms from ordinary CPU memory. Thus, off-device transfer is 3.75–5.08 times the HBM-resident packing cost on this host. Compilation is a negative capability gate: the CPU toolchain lacks an available compiler and the GPU is below the Triton capability floor. No optional serving engine is installed. The result is therefore a tested portable eager baseline and reproducible measurement contract, not a fused-kernel or serving-engine speed claim.

1 Introduction

PRA changes what a model must keep logically available. A long-lived resource may remain addressable through a compact routing representation while only a small selected part of its native K/V participates in one attention operation. That distinction is algorithmically useful, but it leaves a systems question unanswered. A selected set can be small while the runtime still stores every detail tensor in HBM, launches many tiny gathers, copies the same bytes repeatedly, constructs temporary concatenations, or serializes requests around Python control flow. In that case sparsity exists in a trace but not in the resource bill.

This paper asks how to turn PRA into a portable inference primitive before a serving engine becomes semantically aware of PRA memory. The immediate task is not to design a new scheduler. It is to define an execution boundary that can be measured, optimized, and later replaced without changing the meaning of a selected reference.

Three previously separate interfaces must meet at that boundary. The Paper 2 SDK supplies PRAForCausalLM, model-family injection, routing, and generation. Paper 4 supplies authenticated resolvers and the cold, warm, and hot external-memory lifecycle. Paper 6.5 supplies stable typed identities, callable-derived tool schemas, declarative skills, lazy disclosure, and host-authorized tool execution. Paper 7 supplies type-aware compact views, scoped lossless backing, size-gated native indexing, lazy selected-region encoding, selective materialization, and cursors for large results. Paper 8 supplies versioned task state, task-scoped record admission, and durable user/session resolution. We retain these responsibilities and place one orchestration facade above them. This is integration by contract rather than by collapsing distinct policies.

The contributions are:

- a unified runtime contract joining model memory, authenticated resources, typed capabilities and results, durable tasks and sessions, and independently authorized tool execution without collapsing their policy boundaries;
- a physical K/V contract that preserves grouped-query heads, separates selection sparsity, materialization width, and consumer-layer coverage, merges intervals before budgeting, and accounts for transfers, temporary bytes, and bounded cache reuse;
- a corrected source-relative position and request-lifetime implementation, with exact disabled behavior and measured full-native semantic gates on Qwen and Llama plus a localized Gemma global/local topology limitation;
- an engine-neutral, engine-type-aware gateway with independent PRA and ordinary cache/session capabilities, explicit lifecycle, append-only G10 placement, message/resource deltas, affinity hints, and non-sensitive session inspection, while rejecting unsupported phase selection and scheduler hints;
- lazy tools and skills, size-adaptive typed-result indexing, task-scoped acquisition, external-agent bridges, and a user-facing runtime/agent interface; and
- reproducible eager measurements and explicit negative gates for compilation, modern-GPU transfer overlap, and native serving-engine integration.

2 The Systems Gap

2.1 Logical and physical memory are different quantities

Consider L decoder layers. Each reference has T tokens, H_{kv} physical key/value heads, head width D , and b bytes per scalar. Storing complete K/V for the reference requires

$$B_{\text{detail}} = 2LH_{kv}TD b, \quad (1)$$

where the factor two accounts for keys and values. If routing selects S unique tokens, the active payload is

$$B_{\text{active}} = 2L_c H_{kv} S D b, \quad (2)$$

where L_c is the number of memory-consuming layers. The ratio $B_{\text{active}}/B_{\text{detail}}$ is a logical opportunity. It is not peak HBM. Peak HBM also includes model weights, local K/V, routing indices, allocator slack, transfer staging, and temporary packed tensors.

Grouped-query and multi-query attention make the distinction especially important. Warm storage should preserve H_{kv} physical heads. Expanding them to the larger number of query heads before materialization duplicates memory without adding information. The runtime therefore represents every native tensor as

$$K, V \in \mathbb{R}^{B \times H_{kv} \times T \times D} \quad (3)$$

and leaves head expansion to the model-family attention adapter.

These equations require four layer sets, not one overloaded set of “PRA layers.” Let L_A store compact addresses, L_{KV} store detailed native K/V, L_R perform request-time routing, and L_C consume selected memory. The runtime enforces

$$L_C \subseteq L_{KV}, \quad L_R \subseteq L_A \quad \text{for layer-native routing.} \quad (4)$$

The persistent detail term is therefore

$$B_{\text{detail}} = 2|L_{KV}|H_{kv}TD b, \quad (5)$$

whereas one request pays active attention over $|L_C|S$ layer-token states. Address layers may be broader than detail-K/V layers when compact search is useful at several depths. Production defaults do the opposite kind of conservatism: detail K/V is encoded only for the resolved consumer profile, unless profile switching or calibration explicitly requests a union.

2.2 Latency is a sum of visible stages

For one request, PRA overhead is not a single routing timer. We decompose it as

$$t_{\text{PRA}} = t_{\text{lookup}} + t_{\text{graph}} + t_{\text{plan}} + t_{\text{gather}} + t_{\text{transfer}} + t_{\text{pack}} + t_{\text{position}} + t_{\text{metadata}}. \quad (6)$$

Generation and attention follow this preparation. Cold requests also pay source fetch and native encoding; compiled paths pay graph capture and code generation. Reporting only warm attention can therefore reverse the practical conclusion.

3 A Unified Runtime Contract

3.1 One facade, separate authorities

Figure 1 shows the runtime. The facade does not make every component interchangeable in meaning. It gives them one lifecycle and one inspection surface while retaining their ownership boundaries.

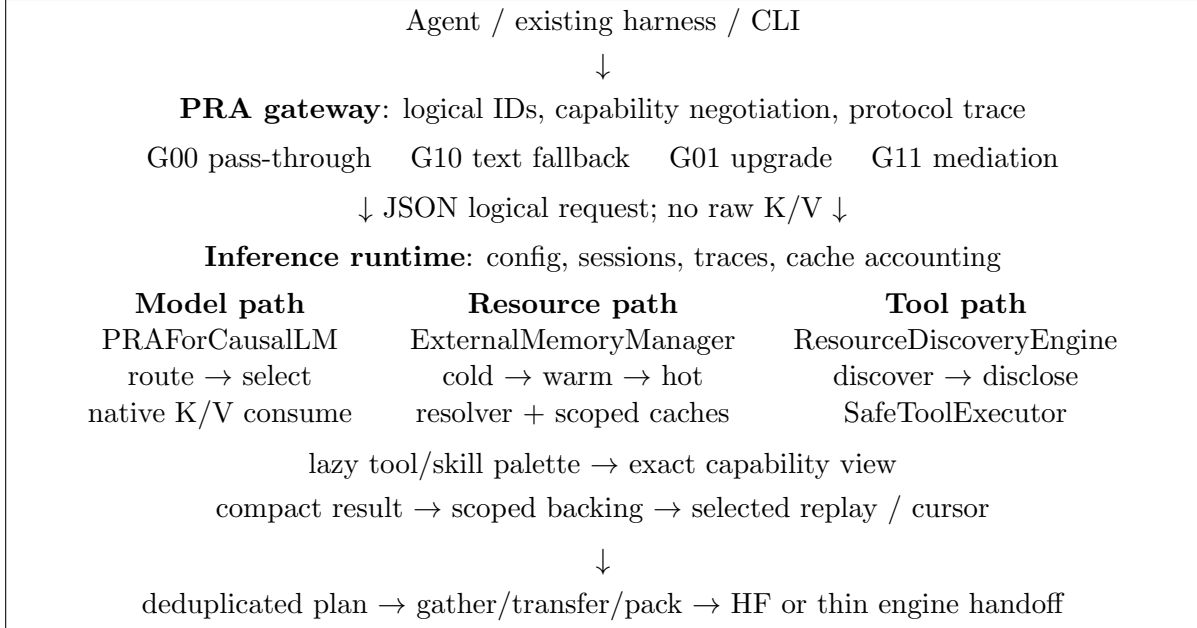


Figure 1: The distributed product boundary. Stable URIs join the paths, but the agent owns task meaning, the gateway owns translation, and the engine owns native K/V. Routing does not authorize source access or tool side effects.

The public configuration has two layers. **PRAConfig** specifies semantic behavior and all four model-layer roles: **address_layers**, **detail_kv_layers**, **routing_layers**, and **consumption_layers**, plus chunking, closure mode, and materialization budget. **PRARuntimeConfig** specifies physical behavior: backend, compilation mode, K/V layout, page size, cache bounds, prefetch policy, and timing synchronization. Both serialize to a versioned JSON artifact. A runtime optimization is valid only when changing the second layer preserves selected identities and output parity under fixed model semantics.

Layer selections use topology-normalized forms: all layers, last n , last fraction, evenly spaced n , or an explicit list. A versioned registry resolves request override, workload/model profile, family balanced fallback, then the all-eligible correctness reference. The named objectives are **reference_correctness**, **quality_max**, **balanced**, and **economy**; the resolved trace records model ID and revision, workload, materialization class, physical layer lists, profile source, and registry version. These are internal optimization objectives. The product registry labels smoke-only realizations of **quality_max** as **QUALITY_MAX_CANDIDATE**. A profile name is not portable evidence by itself.

Address and detail construction have independent lifecycle states (**NOT_BUILT**, **PARTIAL**, **BUILT**, **SKIPPED**, or **DEFERRED**). The detail policy is minimal, profile union, all layers, or explicit. The address policy is routing-only, all candidate routing layers, external-only, or explicit. If a new consumer requires absent detail K/V, the runtime must re-encode missing layers, fail, or follow an explicitly configured profile downgrade. It never silently substitutes a shallower profile.

3.2 A multi-axis execution policy

PRA does not imply one routing frequency. We represent one request with four orthogonal choices. The *selection stage* is request, prefill/completion phase, or token. The *layer scope* is shared logical identities or an independent selection per active layer. The *materialization scope* is request, phase, layer, or token. The *residency policy* retains a payload for that lifetime or releases it after one layer window. Shared identities never mean shared tensors: chunk C_7 resolves to the native K/V independently encoded at each consuming layer.

The global default is request/shared/request/keep. A model default can override individual fields, and a request override takes final precedence. The resolved policy records field-level provenance. Unsupported combinations raise before generation; there is no implicit downgrade. The engine-neutral **PRASelectionPlan** contains stable URI/chunk identities and row-local spans, never tensors. A layer resolver maps that plan to native payloads, and a separate materialization manager applies physical lifetime and residency.

The HF reference path preserves the old high-level request/shared behavior and the old low-level token/per-layer behavior. It adds request/per-layer by capturing all active layers in one memory-disabled prompt pass, routing each once, and freezing each result. It adds token/shared through a request-owned bridge: layers before the designated routing layer receive no current-token PRA memory; that layer selects once, and later layers resolve the same identities to their own K/V. Phase selection is represented but rejected by HF until a cache-correct prefill-to-decode handoff is implemented.

3.3 Stable resources and authenticated sessions

A stable URI is the common identity for text memory, native K/V, tools, and typed observations. Paper 4’s **PRASession** retains tenant, user, session, cache scope, admitted resources, and warm/hot handles. Credentials remain opaque callbacks passed only to resolvers; they are absent from prompts, snapshots, and artifacts. Cache presence never grants authorization.

3.4 Durable logical sessions and task scope

An interactive agent needs continuity longer than one model process. The runtime therefore separates logical **AgentSessionState** from physical **PRASession**. Logical state is resolved by user and session identity and contains an ordered typed-record stream, a task descriptor, and an idempotent event log. Physical state owns admitted resources, native K/V, and result-store handles. Closing it releases accelerator and resolver state without deleting the conversation or task graph.

SessionService is the persistence boundary. Its in-memory implementation provides thread-safe ephemeral operation. Its local implementation hashes path identities, writes versioned JSON through atomic replacement, and rejects stale optimistic versions. A database can implement the same create, resolve, save, list, and delete contract without changing the model path.

Task events create, link, activate, block, resume, complete, or cancel nodes in a validated DAG. Typed records carry producing-task, consuming-task, relation, and event-sequence provenance. Before ordinary PRA ranking, **TaskScopeSelector** admits the active task, its structural closure, or a bounded adaptive neighborhood. Scope controls which records may compete; record and chunk routing still control which admitted detail is materialized.

3.5 Agent and toolset product surface

Toolset couples a callable to its generated schema, namespace, tenant, tags, and declared side-effect class. Bundles compose through the existing **CapabilitySDK**. The built-in local bundle provides file listing, bounded reads and search, exact writes and replacements, Git status, and workspace-rooted command execution. Path operations reject traversal outside the configured workspace. Shell commands remain write-class actions: workspace selection is not a security sandbox.

PRAAgent assembles model, capability discovery, lazy skill folders, safe execution, compact results, durable sessions, and task scope. A turn persists the user record, selects task-relevant records, discloses a bounded tool palette, generates, validates at most a configured number of tool rounds, stores compact output with exact-backing provenance, and persists the final assistant record. The terminal adds resumable sessions and slash commands for tasks, context, and tools. Write and destructive calls are denied by default and can receive one host confirmation; disclosure alone never grants authority.

Paper 6.5’s tool resources enter through the same runtime but a separate decision path. Discovery can identify a tool. Disclosure can expose its schema. A model can propose a call. Only `SafeToolExecutor`, given a request-scoped `ExecutionAuthorization`, can validate identity, arguments, write authority, and destructive authority. Its result becomes a typed observation resource with producer and call provenance. This separation prevents a high retrieval score from turning into ambient execution privilege.

3.6 Lazy tools and skills

Paper 6.5’s latest capability contract is now part of the facade rather than an adjacent experiment. `AgentConfig` accepts ordinary Python callables, explicit declarative `Skill` objects, and a parent directory containing OpenAI- or Anthropic-style skill folders. Callables are inspected into stable, provider-neutral schemas. Folder loaders parse declarative metadata and instructions but never execute bundled scripts.

The default is a two-stage model-side lifecycle. Discovery and Phase A encode only compact selection views under count and token budgets. Once the model or host chooses a stable record ID, `activate_capability` activates the complete tool schema or skill instructions from immutable local backing. It does not rerun semantic discovery. Encoded views may be cached, while active bytes and resident cached bytes remain separate counters. Eager selection or full-view encoding is an explicit policy override.

3.7 Compact typed result backing

A successful tool output need not return as one flat prompt block. Every open `PRASession` now owns an `AdaptiveContextRuntime` with an exact, hash-verified backing store scoped by tenant and session. The runtime infers tabular, log, graph, terminal, or generic structured shape for tool and API results. Type-specific compactors retain schema, counts, statistics, errors, and bounded representative units while lexical, entity, rare-term, schema, and optional summary addresses remain retrieval-only.

The public operations preserve stable record identity: compact inspection, address search, full or selected materialization, and bounded cursor actions. `execute_tool_and_record` joins authorized execution to this record lifecycle while retaining producer, call, and observation provenance. Closing an ephemeral session removes its backing bytes.

Paper 7’s final result changes the product default. Native PRA is appropriate for typed backing only while its ingestion and resident-state cost stays inside an explicit budget. Above that budget, the record must remain recoverable through cheaper addresses, search, cursors, and bounded lazy native encoding; it must not be truncated and labeled as a complete native index.

Accordingly, `ContextPolicy` exposes `max_native_index_tokens` and `max_native_index_bytes`, plus per-`RecordType` overrides. The byte gate can reject an oversized descriptor before tokenization; the token gate more directly bounds model work when token accounting is available. The reference defaults are 4,096 tokens and 65,536 bytes, but these are deployment policy values rather than claimed universal optima.

Every record has one auditable native-index lifecycle state:

NOT_REQUESTED BUILT SKIPPED.SIZE.LIMIT DEFERRED

The ordinary Hugging Face loader automatically prepares in-budget records. `BUILT` means complete backing passed through the production PRA encoder. `SKIPPED.SIZE.LIMIT` preserves compact and cheap-index recovery. `DEFERRED` is an explicit synchronous lifecycle decision, not a claim of background work. After a cheap selector identifies a bounded row, field, line, item, or chunk,

`encode_result_region_native` authorizes that exact region and creates native K/V only for it. Tear-down removes compact, full, and lazy-region references. A shared multi-tenant model cache is not claimed safe without engine-level partitioning.

3.8 User workflow

The supported workflow is deliberately small:

```
agent = AgentConfig(tools=(lookup,), skills_path="./skills")
context = ContextPolicy(
    max_native_index_tokens=4096,
    max_native_index_bytes=65536,
    record_policies={DB_RESULT: TypeContextPolicy(...)})
runtime = PRARuntime.from_pretrained(
    model_id, runtime_config=config,
    agent_config=agent, context_policy=context)
session = runtime.open_session(...)
palette = runtime.activate_capability_candidates(record_ids)
runtime.activate_capability(selected_id)
result = runtime.execute_tool_and_record(..., session=session)
audit = runtime.result_native_index_audit(
    session, result.record.record_id)
if audit.native_index_state == SKIPPED_SIZE_LIMIT:
    detail = runtime.encode_result_region_native(
        session, result.record.record_id, {"rows": [20, 30]})
trace = runtime.inspect()
```

The same facade still accepts direct references, an `ExternalMemoryManager`, a custom discovery engine, and a safe executor. The executed notebook demonstrates model memory, physical materialization, lazy capabilities, compact results, cursors, size-gated native result routing, and selected-region native promotion with an offline Hugging Face model and pure in-memory tools.

4 Physical Materialization

4.1 Planning before copying

Routing produces logical half-open intervals $I_i = (u_i, \ell_i, a_i, b_i)$, naming URI u_i , layer ℓ_i , and token range $[a_i, b_i)$. The planner groups intervals by URI and layer, sorts them, merges overlaps, and then applies a global token budget. If two requested intervals overlap, their common tokens are copied once. The trace reports requested, unique, and dropped token counts, making deduplication and budget truncation auditable.

The portable materializer slices source tensors, transfers selected slices when needed, and packs them by consumption layer. It reports output bytes, transfer bytes, and temporary bytes. It does not infer peak HBM from these values; CUDA profiling records peak allocator state separately.

4.2 Native-reference semantic gate

Reference K/V is captured at source-relative positions. The direct query must therefore begin strictly after every source key that it may attend to. For independent records r with source lengths L_r , the runtime uses

$$p_{q,j} = \max_r L_r + j, \quad (7)$$

not j and not $\sum_r L_r + j$. The maximum avoids query/source position collisions while preserving each record’s independent coordinate system. Earlier code started $p_{q,0} = 0$, which invalidated full-reference RoPE parity. This was a runtime bug, not a routing or training deficiency.

The corrected path enters selected native K/V and local K/V into the same family-native softmax. It retains grouped-query K/V heads physically, preserves the model’s scaling and causal mask, and keeps the immutable materialization plan active through prefill and every decode call. Compact telemetry records query width, local-cache width, and active native width without retaining attention tensors.

Table 1: HF native-reference correctness ladder. E0 is an offline FP32 mechanism test; E2–E3 reuse the 15-case frozen Qwen3-0.6B Paper 8 ablation.

Gate	Condition	Accuracy	Native tokens	Result
E0	One complete reference; visible/native logits	—	8	5.97×10^{-8} max error
E1	Multiple complete records; boundary/dedup audit	—	144	pass
E2	Selected complete task records; all 28 layers	60.0%	80.2	functional
E3	Paper-3 radius-zero evidence; all 28 layers	0.0%	35.7	insufficient

The bug fix closes the semantic blocker but does not make every sparse geometry adequate. Visible full task scope reaches 73.3%, corrected selected full-record native consumption reaches 60.0%, and sparse late-layer or radius-zero paths remain at 0% on this frozen checkpoint. The first expected token is often ranked correctly even when later generation fails. The remaining boundary is autoregressive composition across materialization depth and layer coverage; training is not proposed as a remedy for a broken native path.

4.3 Cross-model semantic reference gate

We next apply one architecture-neutral protocol to three frozen pretrained models. The exact checkpoints are Qwen/Qwen3-0.6B revision `c1899de2`, unsloth/Llama-3.2-1B revision `9535bd9b`, and google/gemma-3-1b-it revision `dcc83ea8`. The official Meta Llama revision `4e20de36` remained license-gated to the available account; the experiment therefore uses a public weight mirror and records that substitution rather than implying that the official repository was tested. All runs use eager FP16 CUDA on the GTX 950M, one model per process.

Before the semantic cases, each model must preserve logits, hidden states, greedy generation, and the complete dynamic cache exactly with PRA disabled. Reference registration must retain native K/V-head count and head width without query-head expansion, preserve source-relative positions, and remain active for two explicit decode calls. The four measured conditions are a visible prefix, the same complete reference as native K/V, an explicitly selected complete record, and a family-declared sparse layer profile. Three one-token continuations test a capital, color, and numeric code. This small diagnostic localizes integration failures; it is not an answer-quality benchmark.

Table 2: Cross-model HF semantic gate. Selected-full and sparse columns report one-token exact accuracy over three cases. Full-native errors are against the same visible prefix. KV-head fidelity and request-lifetime checks pass for all rows.

Model	Disabled	Full native	Selected full	Sparse profile	KV heads
Qwen3-0.6B	exact	0.0364 max	66.7%	8/28 layers; 66.7%	8/16 preserved
Llama-3.2-1B mirror	exact	0.0234 max	66.7%	8/16 layers; 33.3%	8/32 preserved
Gemma-3-1B	exact	partial; 9.5410 max	100.0%	2/4 global; 100.0%	1/4 preserved

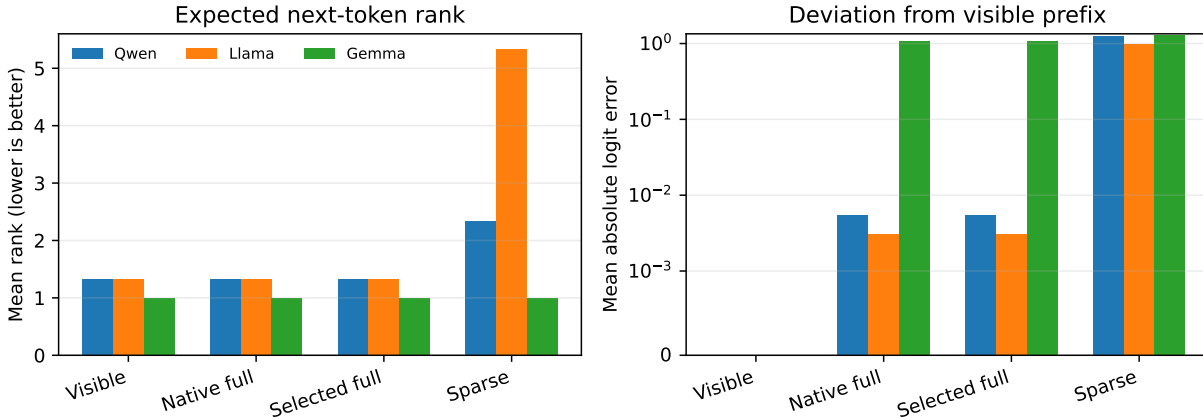


Figure 2: The full and selected-full conditions preserve expected-token rank for Qwen and Llama. Reducing consumer-layer coverage changes semantics despite selecting the same complete record. Gemma’s global-only path retains the top token in these cases but has large logit deviation because unchanged local sliding layers do not consume external K/V.

For Qwen and Llama, full-native replay preserves the visible-prefix top token in all three cases. Maximum FP16 logit error is 0.0364 and 0.0234, respectively. The sparse condition exposes the three independent materialization axes: record selection remains fixed, token width remains the complete selected record, but consumer layers fall from 28 to 8 for Qwen and from 16 to 8 for Llama. Mean expected-token rank consequently moves from 1.33 to 2.33 and 5.33, and Llama accuracy falls from 66.7% to 33.3%. Sparse routing therefore does not establish a semantically adequate consumption profile.

Gemma is a useful failed abstraction boundary. Its 26-layer decoder contains four global-attention layers; the adapter intentionally leaves 22 local sliding-window layers native. Disabled parity, single-head MQA storage, positions, and decode lifetime pass, but global-only native replay differs from the visible prefix by 9.5410 maximum and 1.0667 mean absolute logit error. The one-token top answers remain unchanged, but that coarse agreement is not full-prefix equivalence. We therefore report *partial topology*, not cross-family semantic parity.

4.4 Corrected layer-profile calibration

The earlier sparse column was a diagnostic, not a calibrated deployment profile. We therefore inject all eligible layers once and replay one fixed, complete reference identity through a common profile sweep. Qwen uses all 28 decoder layers; Llama uses all 16; Gemma normalizes over its four eligible

global-attention layers. Candidate profiles are all layers; contiguous late suffixes of 1, 4, 8, 12, 14, 16, 20, and 24 where defined; early and middle blocks; and evenly spaced blocks. Reference width, query offset, native post-position K/V, GQA/MQA head count, and shared-softmax attention remain fixed. The 102 rows comprise three frozen one-token cases per model. They calibrate SDK behavior but do not replace Paper 3’s workload-scale placement study.

Table 3: Corrected fixed-identity layer calibration. ΔLP is target-token log probability relative to the visible-prefix control. Rank is averaged over three cases. “All” means all eligible layers, not necessarily every physical decoder layer for Gemma.

Model	Profile	Consumers	Active layer-tokens	ΔLP	Rank
Qwen3-0.6B	all	28	196.0	+0.001	1.33
	last 24	24	168.0	+0.389	1.00
	last 8	8	56.0	−0.650	2.33
	last 1	1	7.0	−1.705	20.33
Llama-3.2-1B	all	16	106.7	−0.002	1.33
	last 8	8	53.3	−1.986	5.33
	last 1	1	6.7	−2.708	16.67
Gemma-3-1B	all four global	4	28.0	−0.038	1.00
	last global	1	7.0	−7.818	72.33

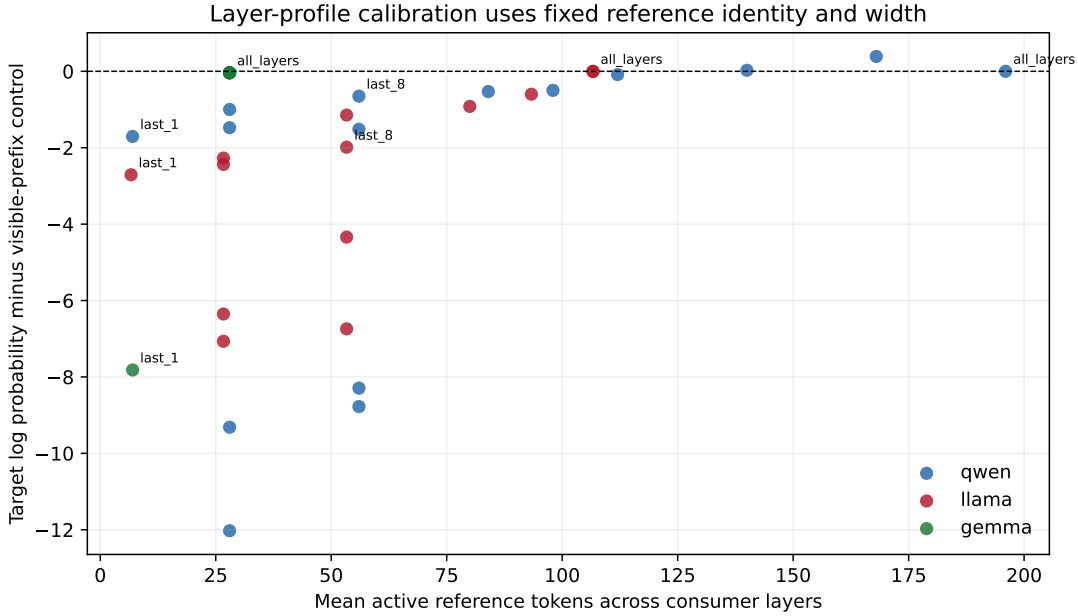


Figure 3: Target-token quality against active layer-token cost with reference identity and token width fixed. The sharp family-dependent trajectories rule out a universal “last eight” product default.

All-layer Qwen and Llama replay are the only tested profiles with maximum full-logit error below .1. Qwen last-24 has the best mean target log probability on these three cases, but its maximum full-logit deviation is 7.14; it is therefore registered under the internal `quality_max` objective but published as `QUALITY_MAX_CANDIDATE`, not as a production-grade `QUALITY_MAX` profile or the balanced product default. Llama requires all 16 layers for the same fidelity criterion. Gemma requires all four eligible

global layers to preserve the tested target tokens, yet still cannot reproduce the hidden computation of its 22 unchanged local layers. Thus **balanced** and **reference_correctness** resolve to all eligible layers for all three families in this registry version. The economy profile is an explicit last-one stress point and carries no quality guarantee.

The storage consequence is intentionally separate. Routing-only address state is retained at one late layer, while detail K/V is retained at the resolved consumer layers. On these short references, Qwen detail storage falls from 802,816 bytes for all 28 layers to 688,128 for last 24 and 229,376 for last 8; the routing index remains 2,048 bytes. A service that needs rapid profile switching may ingest the union of candidate detail layers. The production default instead stores only the chosen profile and re-encodes explicitly if a later request asks for missing detail.

4.5 Product profile and benchmark matrix

The calibrated layer candidates become useful to an SDK only after their semantics and evidence are inspectable. The current registry publishes four profiles: **REFERENCE_CORRECTNESS**, **QUALITY_MAX_CANDIDATE**, **BALANCED**, and **ECONOMY**. The accepted **QUALITY_MAX** name is deliberately reserved and has no evidence row until workload-scale validation supports it. A semantic row is keyed by model ID, immutable revision, workload, and profile. Engine, engine version, hardware, and dtype identify a physical realization of that row; they do not redefine the profile. This separation prevents one GPU measurement from becoming an implied portable latency property.

Table 4 is generated directly from the packaged JSON registry. Quality is mean target-token probability over the same three fixed continuation cases used above. Retention and delta are recomputed against the model’s reference-correctness row. Materialized tokens count selected source tokens, whereas active K/V counts those tokens across consuming layers. Detail savings measure persistent native K/V, so they remain distinct from active attention savings. The compact manuscript table shows active K/V; materialized tokens, detail storage, and the complete physical realization remain in the generated registry and technical table.

Table 4: Model-specific product profiles. This is SMOKE evidence over three fixed cases, not workload-scale quality. Quality-max candidates are explicitly **CALIBRATION PENDING**; Gemma rows also remain partial-topology at the mechanism level.

Model	Product profile	Quality	Ret.	Δ	Active K/V	Save	Evidence	Status
Qwen3-0.6B	Ref. correctness	0.4364	100.0%	+0.0000	196.0	0.0%	Smoke	Measured
Qwen3-0.6B	Quality max candidate	0.4790	109.8%	+0.0427	168.0	14.3%	Smoke	Calibration Pending
Qwen3-0.6B	Balanced	0.4364	100.0%	+0.0000	196.0	0.0%	Smoke	Measured
Qwen3-0.6B	Economy	0.3488	79.9%	-0.0875	7.0	96.4%	Smoke	Measured
Llama-3.2-1B	Ref. correctness	0.5083	100.0%	+0.0000	106.7	0.0%	Smoke	Measured
Llama-3.2-1B	Quality max candidate	0.5083	100.0%	+0.0000	106.7	0.0%	Smoke	Calibration Pending
Llama-3.2-1B	Balanced	0.5083	100.0%	+0.0000	106.7	0.0%	Smoke	Measured
Llama-3.2-1B	Economy	0.1972	38.8%	-0.3111	6.7	93.8%	Smoke	Measured
Gemma-3-1B	Ref. correctness	0.8753	100.0%	+0.0000	28.0	0.0%	Smoke	Partial Topology
Gemma-3-1B	Quality max candidate	0.8753	100.0%	+0.0000	28.0	0.0%	Smoke	Calibration Pending
Gemma-3-1B	Balanced	0.8753	100.0%	+0.0000	28.0	0.0%	Smoke	Partial Topology
Gemma-3-1B	Economy	0.0009	0.1%	-0.8744	7.0	75.0%	Smoke	Partial Topology

Qwen’s last-24 candidate has 109.8% mean-probability retention while reducing active and detail K/V by 14.3%; this narrow positive result motivates its quality-max-candidate label but does not establish general superiority or a production **QUALITY_MAX** profile. Its registry row pairs **evidence_tier=SMOKE** with **profile_status=CALIBRATION_PENDING**. Qwen economy retains 79.9% on the smoke cases with 96.4% K/V savings. The corresponding economy points are poor for Llama (38.8% retention) and unusable for Gemma (0.1%), which is precisely why the SDK does not ship one universal sparse layer rule. Balanced remains all eligible layers for this registry version.

The semantic label resolves to inspectable mechanism settings. Explicit request fields are applied afterward and therefore take precedence over profile defaults. The runtime trace records `profile_requested`, the resolved profile, its source, registry version, evidence tier, measurement status, and separate product-readiness `profile_status`. A request for the reserved `QUALITY_MAX` label or an uncalibrated model/workload pair returns `CALIBRATION_PENDING` rather than borrowing candidate evidence.

Table 5: Resolved technical realization. Counts refer to eligible physical layers. Compression and search are fixed controls in this semantic smoke.

Model	Profile	Compression	Search	Materialization	Address	Detail	Route	Consumer
Qwen3-0.6B	REFERENCE_CORRECTNESS	none_semantic_smoke	fixed_full_reference	full_selected_record	1	28	1	28
Qwen3-0.6B	QUALITY_MAX_CANDIDATE	none_semantic_smoke	fixed_full_reference	full_selected_record	1	24	1	24
Qwen3-0.6B	BALANCED	none_semantic_smoke	fixed_full_reference	full_selected_record	1	28	1	28
Qwen3-0.6B	ECONOMY	none_semantic_smoke	fixed_full_reference	full_selected_record	1	1	1	1
Llama-3.2-1B	REFERENCE_CORRECTNESS	none_semantic_smoke	fixed_full_reference	full_selected_record	1	16	1	16
Llama-3.2-1B	QUALITY_MAX_CANDIDATE	none_semantic_smoke	fixed_full_reference	full_selected_record	1	16	1	16
Llama-3.2-1B	BALANCED	none_semantic_smoke	fixed_full_reference	full_selected_record	1	16	1	16
Llama-3.2-1B	ECONOMY	none_semantic_smoke	fixed_full_reference	full_selected_record	1	1	1	1
Gemma-3-1B	REFERENCE_CORRECTNESS	none_semantic_smoke	fixed_full_reference	full_selected_record	1	4	1	4
Gemma-3-1B	QUALITY_MAX_CANDIDATE	none_semantic_smoke	fixed_full_reference	full_selected_record	1	4	1	4
Gemma-3-1B	BALANCED	none_semantic_smoke	fixed_full_reference	full_selected_record	1	4	1	4
Gemma-3-1B	ECONOMY	none_semantic_smoke	fixed_full_reference	full_selected_record	1	1	1	1

Runtime fields use an explicit missingness contract. The registry contains TTFT, inter-token latency, tokens/s, p50/p95/p99, throughput, peak HBM and RAM, H2D bytes and time, cache hit rate, index build time, prefetch overlap, and concurrency. None was measured by the layer-quality calibration, so the generated view says NOT MEASURED; it does not reuse the separate microbenchmark later in this paper.

Table 6: Physical realization fields for balanced semantic rows. Missing serving measurements remain explicit.

Model	Engine	Hardware	dtype	TTFT	tok/s	p95	Runtime status
Qwen3-0.6B	huggingface_eager	NVIDIA GeForce GTX 950M	torch.float16	NOT MEASURED	NOT MEASURED	NOT MEASURED	NOT_MEASURED
Llama-3.2-1B	huggingface_eager	NVIDIA GeForce GTX 950M	torch.float16	NOT MEASURED	NOT MEASURED	NOT MEASURED	NOT_MEASURED
Gemma-3-1B	huggingface_eager	NVIDIA GeForce GTX 950M	torch.float16	NOT MEASURED	NOT MEASURED	NOT MEASURED	NOT_MEASURED

4.6 Inherited typed-record product policy

The runtime also forward-ports Paper 7’s finalized typed-record policy without reinterpreting its metrics as layer-profile evidence. Per-record compression accepts target-token, maximum-token, and ratio bounds. Typed postings, indexed BM25, and a small embedding index are built independently of the full-body native Q/K size gate. Above that gate, cheap hybrid discovery returns an exact selector; only the selected region is eligible for lazy native encoding and detail-K/V materialization.

In Paper 7’s controlled protocol, type-aware compression reduced mean visible tokens from 180.8 to 54.1, a compact ratio of 25%. Automatic recovery was 71.9%, and cheap-hybrid recall at four was 87.5% over 24 cases. These are inherited measurements with their original cohort and provenance, not new Paper 4.5 quality results. Their role here is to define canonical runtime behavior: full native indexing is budgeted, while cheaper address modes and exact backing identities remain available above the gate.

The task smoke invokes the inherited Paper 8 task graph, provenance attachment, and task-structural selector rather than copying expected IDs into the report. It selects the expected record in all 18 model/condition rows (three tasks, three models, visible and selected-full conditions). Selected-full one-token accuracy equals visible-prefix accuracy for every model. Tool and skill disclosure, safe execution, compact results, and session persistence are shared control-plane contracts and pass the

executed SDK tests and notebook; we do not label them model-generated tool-use results. The base Llama mirror has no chat template, so no Llama tool-call claim is made.

4.7 Caching and reload amplification

The runtime includes a thread-safe, byte-bounded LRU for opaque warm or hot payloads. Besides hits, misses, and evictions, it records loaded and reused bytes. We define reload amplification as

$$A_{\text{reload}} = \frac{B_{\text{loaded}}}{\max(B_{\text{resident}}, 1)}. \quad (8)$$

An apparently high hit rate can still be costly if the misses repeatedly load large references. Conversely, byte reuse can be high even when object-level hit rate is modest. Both views are needed. Native reusable keys include tenant, user, session, resource, layer, payload variant, source revision, positional frame, materialization layout, and scope signature. Raw backing can be reused across compatible encodings, but per-layer, rebound, and packed payloads require exact geometry identity. The audit accepts only the identical key and rejects changed source, position, packing, or request scope. Per-tenant byte caps evict within that tenant before the global cap is considered, and scope-local cleanup cannot evict another tenant’s entries. A concurrent same-URI fixture confirms tenant isolation under pressure and cancellation cleanup. This is logical/runtime isolation, not hard GPU partitioning.

5 Compilation and Engine Boundaries

5.1 Agent, gateway, and engine are independent

The distributed contract has three authorities. The agent owns user-visible task and session semantics, typed records, provenance, and authorization references. The gateway owns transport sessions, logical resource mapping, protocol translation, and explicit capability negotiation. The inference engine owns tensor layout, physical K/V blocks, device placement, attention, and scheduling. The normal wire carries JSON logical identities rather than raw K/V tensors.

The implemented `PRAWireRequest` preserves messages, tenant/session/task IDs, resources, facets, budgets, execution-policy hints, provenance, and required capabilities. `PRAEngineAdapter` reports static capabilities, prepares and closes sessions, and offers generation plus streaming. Four deterministic gateway modes cover both sides of the migration: G00 passes ordinary traffic to an ordinary engine; G10 converts explicitly selected resources to labeled text for a PRA-unaware engine; G01 infers typed resources from system and tool-result boundaries before calling a PRA-aware engine; and G11 preserves an explicit PRA request for a PRA-aware engine. G10 is control-plane PRA with text materialization, not native PRA execution.

Capability negotiation is fail-closed. A request that requires native K/V is rejected by an E0 engine unless it explicitly allows text fallback and selects G10. The trace then names the unsupported capabilities, downgrade, selected logical IDs, gateway mode, correlation ID, and whether native K/V was used. Cross-tenant resources and credential-like request metadata are rejected before translation. Relevance still does not grant source or tool authorization. G01 promotes only explicit record boundaries—system records, tool results, attachments, and already typed messages. It does not infer semantic records from arbitrary unstructured user traffic.

We distinguish E0 protocol integration, E1 native PRA attention, E2 a PRA-aware memory runtime, and E3 scheduler-aware batching. The local HF adapter is the E1 semantic reference. The OpenAI-compatible HTTP adapter and standalone `pra gateway serve` process implement E0. SGLang and FreeToken expose compatible HTTP surfaces, so they are reachable at E0; neither has a PRA-native backend in this Paper 4.5 artifact [6, 2]. Companion Papers 6–6.2 separately exercise

vLLM-Metal, SGLang-MLX, and MLX-LM at E0; those black-box serving runs do not promote this runtime to E2. The portable HF adapter implements E1 native execution and streaming but remains stateless between gateway calls. The configurable remote E1 envelope carries incremental messages and resource operations when the selected engine advertises those capabilities. The HTTP gateway emits OpenAI-compatible server-sent events. Cancellation is cooperative at token boundaries; request-owned references are released only after the decode worker stops. No backing content is included in trace metadata. The capability object names native K/V, logical references, streaming, selected-interval materialization, request lifetime, host/device residency, tenant isolation, phase selection, and scheduler hints independently. The HF reference advertises the first seven supported properties and explicitly rejects phase selection and scheduler hints. This prevents an E1 adapter from being mistaken for an E3 scheduler merely because both accept the same logical wire request.

5.2 Session-Aware Gateway and Conventional Prefix-K/V Reuse

PRA does not replace ordinary prefix caching. A session may simultaneously retain a conventional sequential-prefix K/V cache and separately address PRA records or native detail. The gateway preserves append-only prefixes for ordinary engines and can send logical deltas to PRA-aware remote engines. Figure 4 makes these state classes explicit. The inference engine physically owns the ordinary prefix cache. PRA logically owns resource identities, versions, addresses, and selection; the physical owner of their indexes or selected native K/V depends on E1–E3 integration. A hit or invalidation on one axis therefore says nothing by itself about the other.

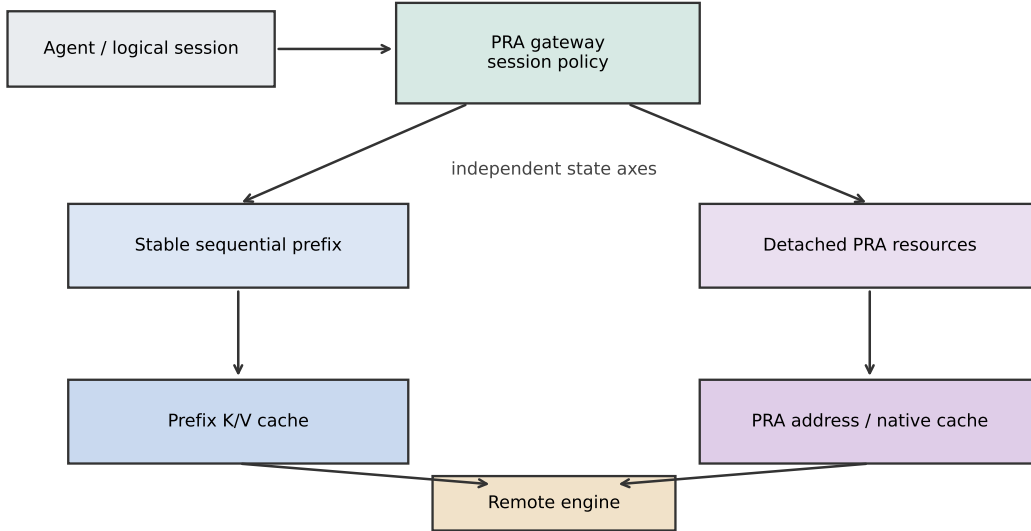


Figure 4: The session-aware gateway maintains two independent state axes. Sequential prompt K/V follows the engine’s ordinary prefix-cache policy; detached PRA resources follow the PRA lifecycle and may be text-materialized at E0 or retained natively at E1+.

The wire contract promotes `history_mode`, engine-session identity, prefix handle, cache-affinity key, maximum decode length, and resource operations to typed fields. `FULL` means that the request supplies canonical complete history; `DELTA` means only appended messages; `AUTO` resolves from capabilities rather than message count. Stable resources similarly produce `ADD`, `UPDATE`, `REMOVE`, or `UNCHANGED` operations while preserving ID, URI, version digest, record type, and authorization scope.

A versioned engine registry resolves OpenAI-generic, vLLM, SGLang, FreeToken, Ollama, Hugging Face, llama.cpp, MLX, and custom profiles. Unknown compatible servers safely remain E0, full-history, and cache-telemetry-unknown. vLLM and SGLang profiles represent automatic prefix-cache capability but do not infer native PRA from engine availability or from the companion E0 smoke runs. The local HF reference remains E1-native but stateless across calls; the session/resource-delta row is exercised by a deterministic remote-engine contract fixture. Explicit configuration may override registry defaults; resolved capabilities, rather than labels, control transport.

For G10, the former message-zero synthetic system block remains only as a legacy experimental control. The default inserts selected evidence into the current user turn, after the byte-identical prior serialized conversation. Role-sequence tests cover Qwen, Llama, Gemma, and generic OpenAI-style transports. This placement preserves template validity and the previous prefix. It does not imply that changing evidence is cacheable: only the prior sequential prefix is reusable.

The sidecar is keyed by tenant, logical session, and model. It retains only ephemeral engine handles, prefix fingerprints, resource versions, policy digests, and affinity hints; durable canonical history continues to use `SessionService`. Preparation is idempotent, explicit close releases engine state, and cancellation of one stream does not close its reusable session. Model revision, chat-template, visible-prefix, history-rewrite, engine-restart, and invalid-handle changes invalidate the relevant handle. Updating one detached resource emits a resource delta without invalidating an otherwise stable conversation prefix. The cache-affinity key is only a future scheduler hint; no worker-placement result is claimed.

5.3 Agent-side compatibility before native engine work

The pre-SGLang deployment gate asks whether PRA’s logical record semantics can survive an existing agent without patching its model runtime. The `DeepSeekHarnessPRAAdapter` consumes durable tool-result, session-reference, and attachment events. The `PiCodingAgentPRAAdapter` consumes Pi’s public tool-completion and completed-message events. Both deduplicate event identities, retain typed provenance and task/session IDs, and emit the same tensor-free `PRAWireRequest` used by the HTTP gateway.

Across 10 deterministic five-seed contract cases, the DeepSeek and Pi bridges each pass 100.0% and 100.0% of identity, typing, task-metadata, deduplication, and explicit-fallback checks. In every G10 case the ordinary engine receives labeled evidence text, zero logical resources, and no native-K/V claim. These tests validate the integration boundary, not the quality or latency of either upstream agent.

5.4 Compilation ladder

The eager path is the correctness baseline. The first compiled primitive is a paired `index_select` over K and V with dynamic selected-token count. Cold compile time and warm latency are separate rows, and parity is checked against eager outputs. Failure never falls back into an eager row.

Triton, custom CUDA, and fusion follow profiling rather than precede it. The candidate hot paths are interval gather, deduplication/remap, packing, and position preparation. A kernel is justified only if it improves the complete request after launch, transfer, and synchronization costs. This ordering prevents a fast isolated kernel from becoming a misleading end-to-end result.

5.5 Thin vLLM handoff

PagedAttention showed how virtualized K/V blocks improve serving utilization and batching [4]. Deep PRA integration would additionally make logical PRA blocks, semantic residency, and retrieval-aware prefetch visible to the scheduler. That is outside this paper’s boundary.

The implemented `VLLMThinBackend` instead emits a `VLLMThinRequest`: prompt, stable selected URIs, materialized token count, and ordinary K/V metadata. Semantic scores do not cross the boundary. An ordinary engine can consume already selected memory without owning retrieval policy. When no executor is installed, the backend produces the handoff for inspection and raises on generation rather than pretending to serve.

6 Relation to Existing Systems

PRA is complementary to several kinds of memory optimization. H2O and SnapKV select or retain influential cache entries inside an active sequence [8, 5]. PRA begins from stable external identities and query-time addressability, so its inactive detail K/V may live outside the accelerator. These methods can eventually share page layouts or eviction mechanisms, but their selection semantics differ.

Prompt Cache and CacheBlend reuse precomputed prompt or retrieved-knowledge state [3, 7]. Their lesson is directly relevant: recomputation and transfer can dominate. PRA adds bounded semantic selection before native K/V consumption. RetrievalAttention similarly joins vector retrieval with long-context attention, reinforcing the need to count index and transfer work rather than only selected attention work.

GQA reduces the number of physical K/V heads [1]. Our runtime preserves this layout through warm storage and materialization. It does not expand K/V to query-head count as an implementation convenience.

7 Experimental Design

7.1 Questions and claim levels

The initial gate asks four narrow questions. First, does the portable selected- K/V implementation preserve exact tensor parity? Second, how large are eager gather and interval-pack costs? Third, how much does CPU residency add for the same selected payload? Fourth, can the local toolchain execute the compiled path? Quality and routing selection remain frozen.

Claims are labeled by evidence. *Measured* means executed on the recorded host. *Inherited* means read from a checked-in earlier experiment with its own protocol. *Contract only* means code exists but no external engine ran. *Architectural* means only an interface comparison is made.

7.2 Gateway session and prefix protocol

The controlled gateway fixture executes five turns with changing selected records under four conditions: legacy message-zero G10 insertion, prefix-preserving G10, E0 message deltas, and an E1 logical message/resource-delta contract. It records whether each prior serialized message sequence remains an exact prefix, the reusable-message fraction, invalidations, JSON message and resource bytes, delta bytes, and session reuse. The workload is deterministic and uses a simulated adapter so that transport semantics can be isolated from model quality and network variance. Byte counts refer to the fixture’s JSON envelopes, not tokenizer output or K/V allocation. Physical cache hits remain unknown unless adapter telemetry supplies them.

7.3 Portable primitive protocol

The new benchmark uses PyTorch 2.12.1 with CUDA 12.6 on Windows 10, Python 3.10.11, and an NVIDIA GeForce GTX 950M with compute capability 5.0. Each synthetic tensor has four physical K/V heads, width 64, and 8,192 candidate tokens in float32. Selection admits 512 tokens (6.25%).

Batch sizes are one and four. CPU rows use five warmups and 30 timed repetitions; CUDA rows use ten warmups and 50 repetitions. Every CUDA timing synchronizes before and after the operation.

Indexed gather uses evenly distributed selected token IDs. Interval packing uses eight intervals under the same 512-token budget. The hierarchy comparison holds intervals and output bytes fixed while moving sources among HBM, pinned CPU, and ordinary CPU. Compiler errors are artifact rows. The benchmark is a mechanism profile, not TTFT or throughput for a language model.

The layout sweep holds 8,192 total source tokens across four references and two layers. It packs 32-token pages in layer-major, reference-major, chunk-major, or block-major order, then restores an identical 512-token logical selection. Store construction is a cold cost; reported layout gather latency follows the same warmup and repetition protocol as the other device rows.

7.4 Inherited profiles

Two earlier artifacts answer adjacent questions. The Paper 2 public-API demo ran frozen Qwen3-0.6B over two QASPER and two HotpotQA examples. It records reference ingest, query encoding, routing, generation, K/V fractions, transfers, and peak GPU allocation. The Paper 4 lifecycle fixture ran eight deterministic resources through authenticated cold, warm, and hot states. Its answer proxy is required-document availability, not language quality. We do not pool these measurements with the new microbenchmark or compute cross-host speedups.

7.5 Execution-policy mechanism profile

We instantiate the same random-weight, three-layer tiny Llama configuration for five seeds and ingest two stable synthetic references. Every condition uses three active PRA layers, deterministic decoding, and at most three generated tokens. We compare request/shared at first and last routing layers, request/per-layer, token/shared at first and last routing layers, and token/per-layer. The profile records routing and selection epochs, temporal and layer Jaccard, selected-identity agreement, and generation wall time. It tests execution semantics and relative routing work, not language quality or production latency. EOS can terminate before three tokens, so token-level operation counts are averaged over the actual forwards.

8 Results

8.1 Stable prefix and session-delta contract

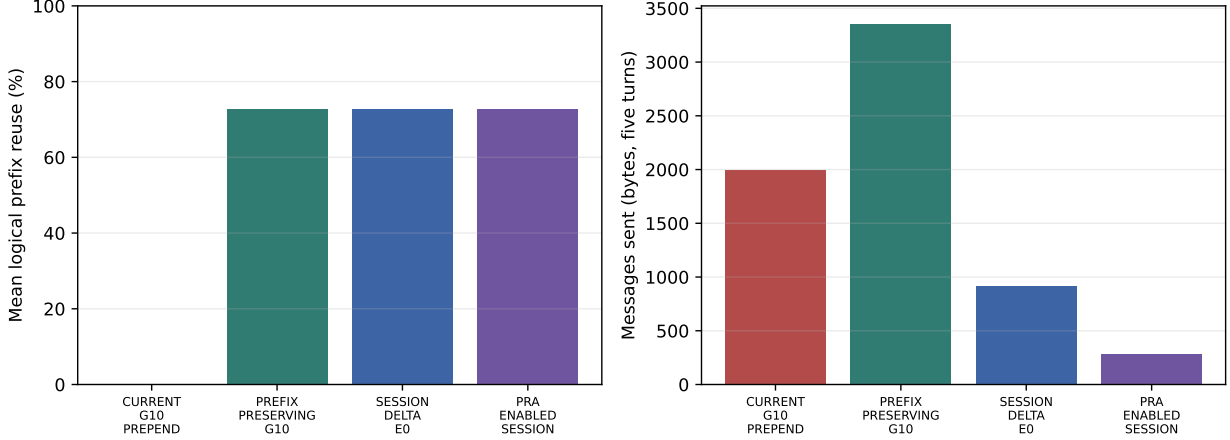


Figure 5: Five-turn deterministic gateway profile. The left panel reports logical reuse of the exact prior serialized message prefix after turn one; the right reports message bytes sent. Resource bytes are reported separately in Table 7.

Table 7: Controlled gateway session profile. Stable turns use the four turns after initial setup. Reuse is logical serialized-message reuse, not an engine-reported cache hit.

Condition	Stable turns	Reuse	Invalidations	Msg bytes	Resource bytes
CURRENT_G10_PREPEND	0/4	0.0%	4	1995	0
PREFIX_PRESERVING_G10	4/4	72.8%	0	3355	0
SESSION_DELTA_E0	4/4	72.8%	0	921	0
PRA_ENABLED_SESSION	4/4	72.8%	0	281	2297

The legacy message-zero G10 control changes the leading evidence block on every eligible turn, preserving zero of four prior prefixes and recording four invalidations. Moving evidence to the current turn preserves all four prior prefixes, with a mean logical reuse fraction of 72.8%. Full-history prefix-preserving G10 sends 3,355 message bytes because it still retransmits history. The session-delta E0 contract retains the same logical prefix while reducing message transport to 921 bytes; four turns reuse the prepared engine session.

The E1 contract sends 281 message bytes and 2,297 resource-envelope bytes, using resource additions, updates, and removals rather than repeating resource contents indiscriminately. Its combined delta bytes are not lower than the tiny text fixture because typed resource metadata and operation envelopes are explicit. This is a semantic and accounting result, not evidence that PRA transport is always smaller. All four conditions report physical prefix-cache hits as unknown. No vLLM/SGLang affinity, TTFT, throughput, or scheduler result is inferred from logical prefix stability.

8.2 Selection sharing removes repeated routing work

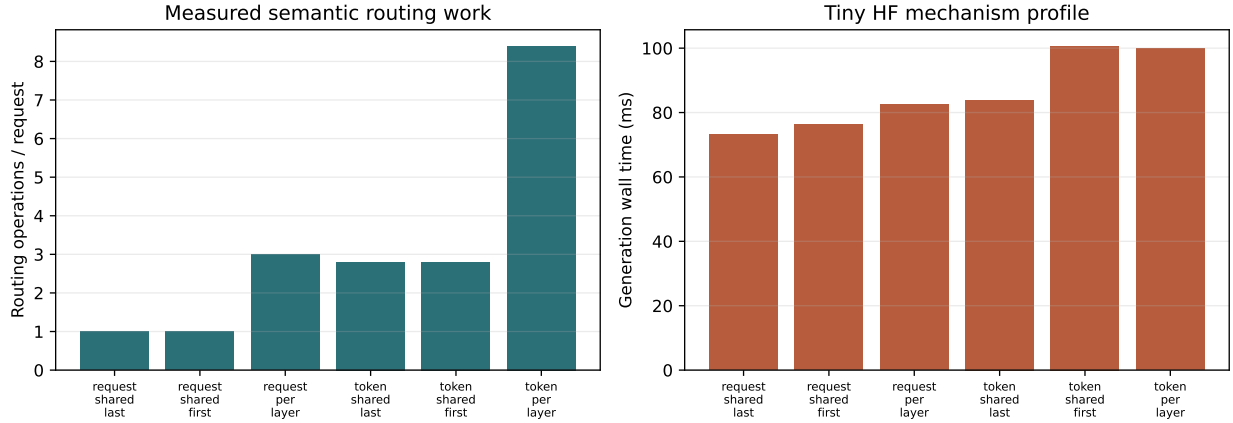


Figure 6: Five-seed tiny-HF mechanism profile. Routing operations validate the execution contract; wall time is host- and model-specific and is not a serving speed claim.

Table 8: Implemented policy points in the five-seed mechanism profile. Times are means over random-weight tiny models and do not measure answer quality.

Condition	Routing operations	Generation ms
Request/shared, last	1.0	73.1
Request/per-layer	3.0	82.6
Token/shared, first	2.8	100.7
Token/per-layer	8.4	100.0

Request/per-layer performs exactly three routing decisions, but amortizes them over generation. Token/shared and token/per-layer observe the same average number of autoregressive forwards; sharing reduces routing work by $3.0\times$, from 8.4 to 2.8 operations. Tiny-model wall times are effectively tied for the two token modes, which means Python and model overhead dominate at this scale. The routing-layer sweep also changed selected identities, confirming that routing layer is a semantic parameter rather than a scheduling-only knob. The stable selections within these short completions yield temporal Jaccard one; longer trained-model studies are needed before making a churn or quality claim.

8.3 Portable eager primitives

Table 9: Median warm latency in milliseconds. All measured outputs match the eager reference exactly. Each output contains 512 selected tokens per request.

Device	Primitive	Batch 1	Batch 4
CPU	indexed gather	0.133	0.835
CPU	interval pack	0.228	0.938
CUDA	indexed gather	0.253	0.478
CUDA	interval pack	0.415	0.613

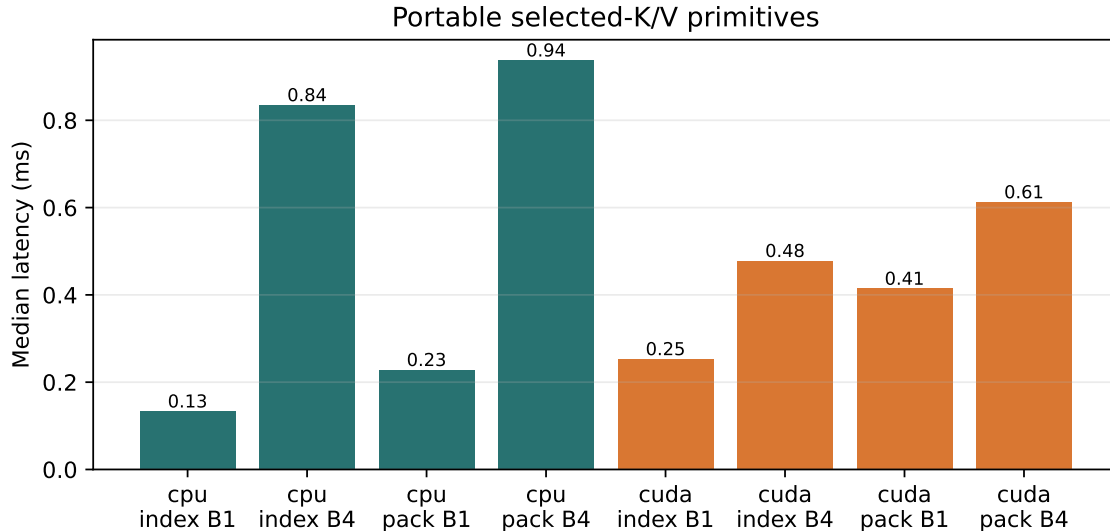


Figure 7: Portable primitive latency. This older GPU does not make small batch-one gathers faster than the CPU, but CUDA scales better at batch four.

The indexed CUDA path is 0.253 ms at batch one. The eight-interval pack is 0.415 ms because it launches and concatenates multiple slices. At batch four, CUDA indexed throughput increases to roughly 4.28 million selected tokens/s, whereas the CPU reaches about 2.45 million million. These values identify packing and fragmentation as optimization targets; they do not include routing or attention.

8.4 Residency dominates the selected payload

Table 10: Median CUDA interval materialization latency for an identical selected payload. Output size is 1 MiB at batch one and 4 MiB at batch four.

Source residency	Batch 1 (ms)	Batch 4 (ms)
HBM resident	0.424	3.371
Pinned CPU → CUDA	1.592	6.655
Ordinary CPU → CUDA	2.157	8.660

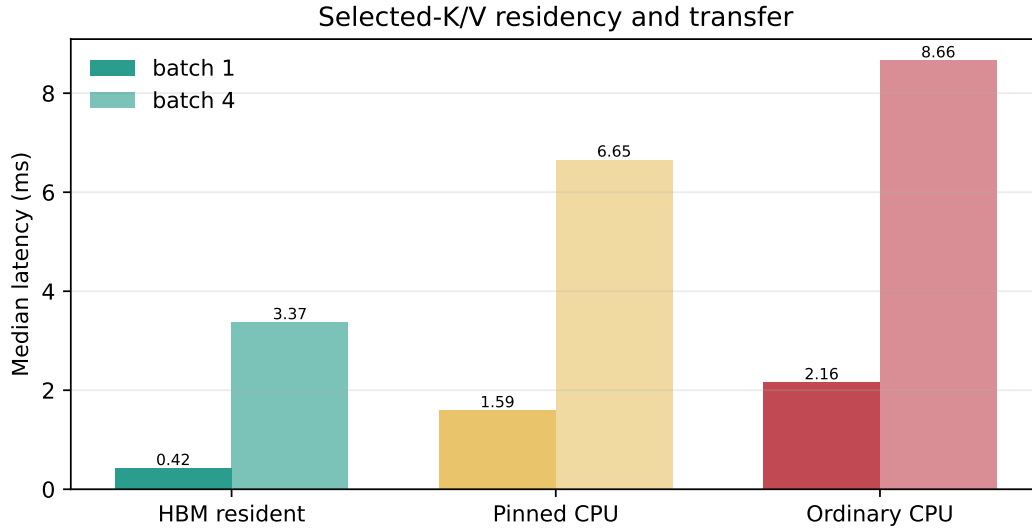


Figure 8: The same logical selection has different physical cost depending on residency. Pinned transfer helps, but does not erase the hierarchy boundary.

At batch one, pinned transfer is 3.75 times the HBM-resident cost, and ordinary CPU transfer is 5.08 times that cost. This is the central result of the initial systems gate. A configuration that moves inactive K/V off accelerator reduces residency, but request latency then depends on locality, cache reuse, and transfer overlap. Any future headline must report both memory and latency.

8.5 Physical layout is secondary to fragmented remapping

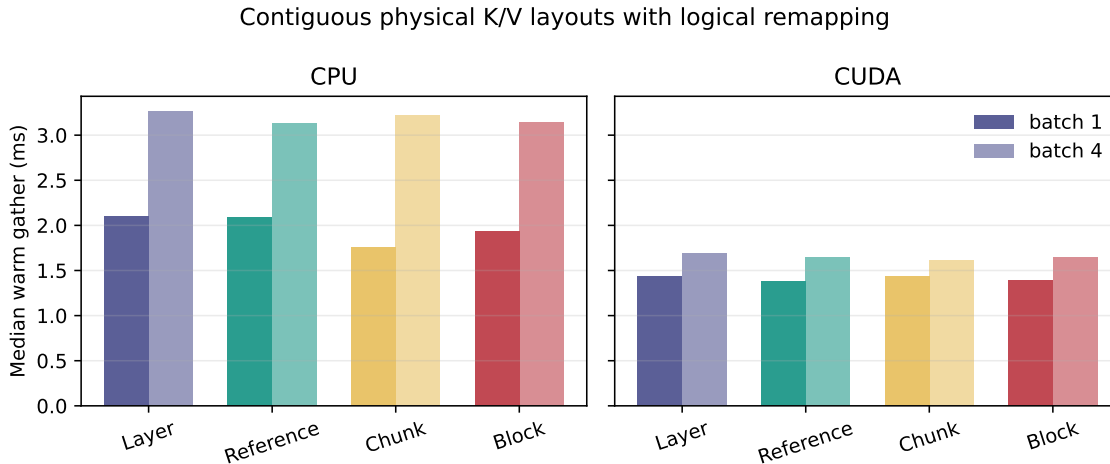


Figure 9: Warm materialization from four physically distinct contiguous stores. Every layout restores the same logical per-layer K/V before attention.

Layer-major, reference-major, chunk-major, and block-major stores place the same four-reference, two-layer native K/V in different contiguous orders. A compact placement index maps URI, layer, and logical page offsets back to physical ranges. All outputs match the unpacked reference exactly. On CUDA, the slowest and fastest medians differ by only 3.8% at batch one and 5.0% at batch four. The

nominal fastest layouts are reference-major and chunk-major, respectively, but those small differences do not justify a universal preference on this synthetic cohort.

More importantly, all layout-remap paths take roughly 1.4–1.7 ms on CUDA, several times the single-source interval pack. They reconstruct selected logical ranges across Python placement records and page fragments. The next optimization target is therefore indexed vectorized remapping and fewer launches, not a claim that one physical ordering has won.

8.6 Compilation is a negative capability gate

The `torch.compile` API is present, but neither local compiled path is valid. CPU Inductor cannot find the Microsoft C/C++ compiler. CUDA compilation rejects compute capability 5.0 because the Triton backend requires capability 7.0 or newer. No warm compiled latency is reported and no eager result is relabeled as compiled. Triton and custom CUDA therefore remain untested.

8.7 Earlier profiles locate larger costs

The new primitive measurements explain the cost of moving an already selected payload. Earlier end-to-end artifacts provide the complementary scale: how much time was spent before and after materialization when the public model API and authenticated resource lifecycle were exercised.

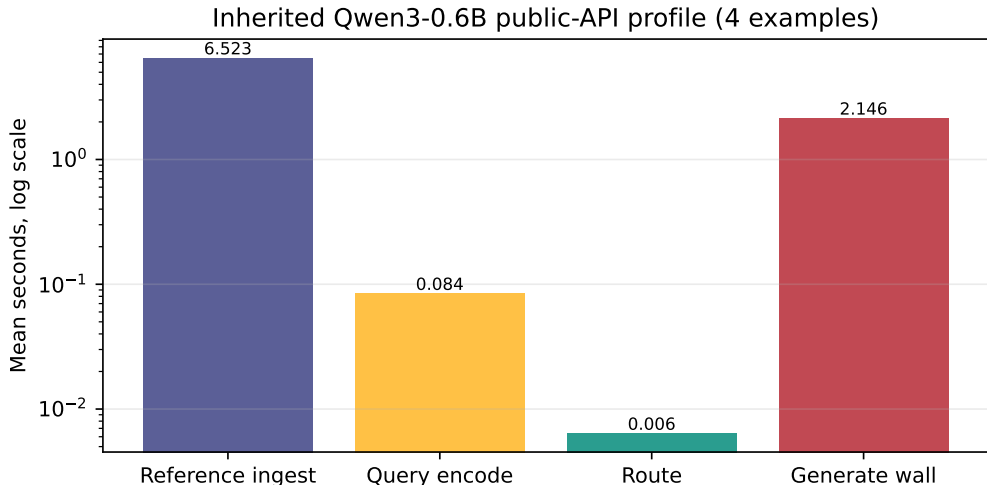


Figure 10: Inherited Paper 2 Qwen3-0.6B profile over four examples. The log scale shows that routing time is small relative to native reference encoding and generation on that protocol.

The inherited profile reports 6.40 ms mean routing time, 6.52 s mean reference ingest, and 2.15 s mean routed generation wall time. Mean evidence recall is 0.667 and materialized detail K/V is 0.066 of the candidate total. This supports optimizing cache reuse and encoding amortization before replacing the router with a custom kernel. The cohort contains only four examples, so it does not establish a general workload distribution.

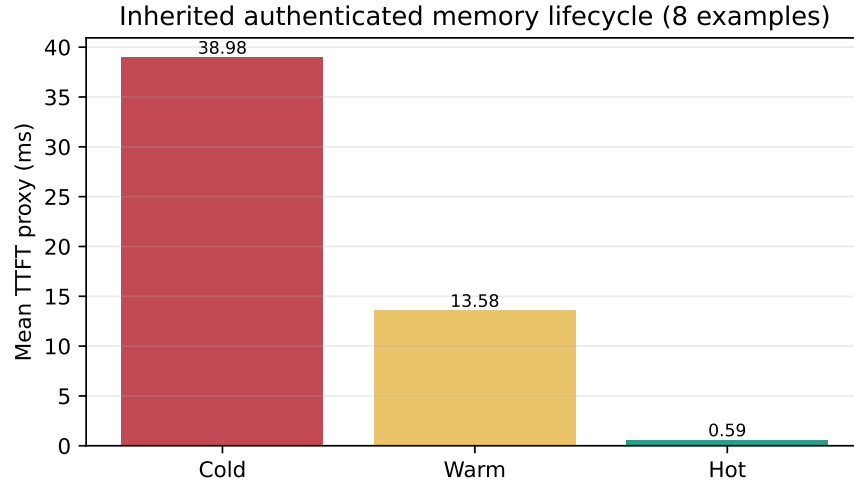


Figure 11: Inherited Paper 4 lifecycle proxy over eight examples. Cold includes fetch and native encoding; warm reuses native encoding; hot reuses selected K/V.

The lifecycle TTFT proxy falls from 38.98 ms cold to 13.58 ms warm and 0.59 ms hot. Required-document recall is 0.875. The result validates state transitions and shows the size of the reuse opportunity, but it is not an LM answer-quality or production-serving measurement.

9 Compatibility and Product Boundary

Table 11: Implementation status on the measured branch and host.

Runtime	Status	Boundary
HF/PyTorch eager	measured	Native PRA model API and portable K/V primitives
Standalone gateway	contract tested	G00/G10/G01/G11, session lifecycle, prefix-preserving fallback, deltas, debug/close API, and HF-backed SSE
OpenAI-compatible HTTP	E0 implemented	Full-history safe default; typed engine profiles and explicit text fallback
DeepSeek/Pi agent bridges	contract tested	Typed event/RPC capture and explicit G10 fallback
<code>torch.compile</code>	blocked on host	API implemented; compiler/GPU gates recorded
Triton/custom CUDA	not run	Requires supported GPU and profiling evidence
vLLM thin	contract only	Scheduler-unaware identity and K/V handoff; E0 engine smoke is reported separately in Paper 6
SGLang/FreeToken	E0 contract	Compatible HTTP; SGLang E0 smoke is reported separately in Paper 6.1; no native PRA backend
TensorRT-LLM/MLX	architectural only	MLX E0 and rotating-cache controls are reported separately in Paper 6.2; no Paper 4.5 engine claim

The product surface includes installable package metadata, Hugging Face loading, runtime config serialization, direct and external reference registration, session isolation, callable and folder-backed capability registration, lazy view activation, typed resource discovery, safe execution, compact result ingestion, selected replay, address search, cursors, size-adaptive native result routing, lazy native-region promotion, generation, inspection, benchmark commands, and two executed notebooks. The CLI exposes `runtime init`, `inspect`, `capabilities`, `benchmark`, `prepare-vllm`, and a persistent `agent chat` terminal. The `pra gateway serve` and `pra-hf gateway serve` commands launch the same independently testable gateway with health, capability, logical-request, and OpenAI-compatible endpoints. A deployment can therefore discover unsupported paths before serving.

10 Limitations and Next Gates

The current evidence is intentionally incomplete. The GPU is old, the benchmark is synthetic, and no full serving engine is installed. We have not measured continuous batching, request-level p95/p99, asynchronous transfer overlap, a production page allocator, prefetch, multi-session contention, or fused attention preparation. The LRU fixture validates byte accounting but is not a production cache policy study. The inherited language-model cohort is too small for quality claims. The new capability and compact-result workflows and Paper 8 task/session integration are executed contract tests, not a new latency or answer-quality experiment. The shell is intentionally single-agent; subagents and learned long-term memory remain scoped to Papers 9 and 10. Model-resident result routing currently requires an isolated model session; safe concurrent multi-tenant partitioning of model-resident result K/V remains an engine-level gate. The multi-axis HF bridge serializes high-level generation on one wrapped model to isolate temporary adapter state. It is a correctness reference, not a continuous-batching implementation. The cache key now rejects physical reuse across changed source revisions, positions, materialization layouts, or scopes, and a concurrent fixture covers logical tenant isolation, pressure, and cancellation. It does not establish hard GPU partitioning or production multi-tenant throughput. Phase selection and E2/E3 engine integration remain open. The G01 parser recognizes explicit system, tool, and attachment boundaries; it does not claim semantic recovery from arbitrary opaque traffic.

The Paper 4.5 gateway profile is a deterministic transport experiment. It does not use a real serving engine or distributed scheduler, and therefore does not measure physical prefix-cache hits, worker affinity, warm-prefix TTFT, continuous batching, index residency, or network cost. The reported reuse fraction counts exact serialized messages, not tokenizer tokens. Engine profiles are conservative defaults and require runtime probing or explicit configuration before stronger capabilities may be advertised. Companion E0 engine studies do not change this profile’s evidence tier and are intentionally kept outside Paper 4.5’s HF-centered metric tables.

The cross-model cohort contains only three one-token continuations per model. It is sufficient to expose exact disabled behavior, K/V topology, position, lifetime, and coarse semantic drift, but not to estimate broad language quality. The Llama row uses a public weight mirror because the pinned official repository was inaccessible. Gemma is partial rather than passed: its global-only adapter cannot make external K/V visible to unchanged local sliding layers. Sparse profiles are declared per family and are not assumed to preserve quality. The layer calibration reuses that diagnostic cohort. Its 102 rows vary profiles, not independent examples, so they identify gross transport drift and set conservative SDK defaults but cannot estimate downstream answer quality. Qwen last-24’s target-token gain is especially vulnerable to this small-cohort selection effect. It is consequently exposed only as `QUALITY_MAX_CANDIDATE` with pending calibration, not as `QUALITY_MAX`. Paper 3 owns retrospective placement evidence on MuSiQue and 2Wiki; a future calibration should add held-out QASPER, HotpotQA, typed-result, and long-generation workloads before any sparse profile becomes a default.

The next experiment should move to a compute-capability 8.0 or newer GPU with a pinned software

stack. It should first reproduce eager parity, then measure `torch.compile` cold and warm paths. Only measured hot spots should move to Triton. Layout and hierarchy sweeps should cross contiguous versus indexed selection, whole chunks versus intervals, GQA head count, batch heterogeneity, and warm-cache locality. End-to-end runs should report TTFT, inter-token latency, throughput, peak HBM, off-device bytes, temporary bytes, transfers, and reload amplification together.

The next policy experiment should use a trained model and retrieval workload, hold selected-token budgets fixed, and compare answer quality, TTFT, inter-token latency, selection churn, and routing-layer disagreement. Phase/shared requires a cache-correct prefill handoff before it enters that comparison.

Deeper engine integration remains outside Paper 4.5. Companion engine papers have now established reproducible E0 baselines; their next step remains thin: external routing, selected native-K/V materialization, then ordinary vLLM execution. Disabled-PRA parity, native positions, masks, GQA, and batching must pass before speed comparisons. If useful behavior requires semantic residency, retrieval-aware scheduling, or cross-request PRA block sharing, that work belongs to the deep engine paper.

11 Conclusion

PRA’s selected-memory fraction is an algorithmic fact, not a systems result. This paper turns that distinction into code and measurement. One runtime now connects the public model API, authenticated memory lifecycle, typed resources, lazy tool and skill disclosure, safe execution, compact result backing, selective replay, durable tasks and sessions, physical K/V planning, caching, profiling, and backend handoff. Typed results build full native indexes only within configurable token/byte ingestion budgets; larger records retain cheap recovery interfaces and promote selected regions lazily. Its policy lattice makes routing frequency and physical lifetime inspectable, while the gateway permits ordinary and PRA-aware agents and engines to coexist without sending native tensors over the logical wire. Conventional sequential-prefix K/V and detached PRA resource/native state are now separate capability and lifecycle axes. The tested gateway preserves every eligible prior prefix in its five-turn fixture and sends typed message/resource deltas through a prepared session, while correctly leaving physical cache hits unknown. The HF semantic reference now passes full-prefix logit equivalence on the tested Qwen and Llama checkpoints, plus record-boundary deduplication and request-long decode-lifetime gates. Gemma validates native MQA mechanics but identifies a global/local topology boundary rather than a full semantic pass. Layer placement is now a versioned calibration result rather than an implicit “PRA layer” convention. Separate address, detail-K/V, routing, and consumer sets make persistent storage and request-time computation independently auditable; missing detail can never trigger a silent profile downgrade. The eager primitives are exact and inexpensive in isolation, but off-device transfer multiplies their cost on the measured host. Compilation remains a negative gate, and companion E0 engine runs leave native E2/E3 integration unmeasured. The practical conclusion is conservative: cache locality, physical residency, and context lifecycle deserve attention before kernel novelty, and every future speed claim must match semantics while accounting for the whole request.

A Reimplementation Guide

The stable public entry points live in `src/prahf/runtime.py`. A reimplementation needs five small contracts. First, represent warm K/V as equal-shaped tensors $[B, H_{kv}, T, D]$. Second, express selections as stable URI, layer, and half-open token intervals. Third, merge overlaps before applying a hard global token budget. Fourth, pack selected tensors by consumption layer without expanding grouped-query heads. Fifth, emit stage times and physical bytes independently of routing-quality metrics.

The model adapter only needs `add_reference`, `generate`, and `inspect`. The thin serving request contains selected identities and materialized-token metadata, never semantic scores. Tool execution receives a separate authorization object after discovery. These boundaries allow another runtime to reproduce the interface without adopting the Python implementation.

Paper 8’s finalized geometry adds one further invariant. A frozen selection is an immutable set of source-local anchors. Expansion may change the interval width or choose the complete selected record, but it cannot cross record boundaries or change source identity. Overlap is merged before budgeting, and each consumer resolves the same logical interval to that layer’s own projected K/V. The query starts after the longest source extent, while separate records retain source-relative positions. Raw intervals are retained for diagnostics; the canonical union is the only geometry materialized. The named `paper3_default`, `paper7_selected_detail`, and `paper8_full_record_diagnostic` profiles resolve to ordinary runtime configuration. The public sequence is `route`, `freeze_native_selection`, `plan_native_materialization`, then `generate_with_native_plan`. Task operations follow the analogous control boundary: parse untrusted proposals, validate them against a copy of the task graph, and only then replay accepted events through the session service.

A.1 HF native-reference implementation map

The four-role resolver is in `src/pra_hf/layer_profiles.py:1--430`. Lines 27–77 define independent encoding and missing-detail policies; lines 81–178 normalize all, suffix, fractional, evenly spaced, and explicit profiles; lines 181–244 validate the L_A , L_{KV} , L_R , L_C invariants and serialize provenance. The stable public fields and registry precedence are applied in `src/pra_hf/config.py:19--420`. The versioned model entries live in `src/pra_hf/model_profiles/layer_profile_registry.json`.

Lines 328–510 of the `HF injection.py` module separate reference payloads. Every injected layer can produce compact identity metadata for cross-layer mapping, address layers retain their search representation, and only detail layers retain native K/V. The cache record exposes `has_detail_kv` in `src/pra_torch/memory.py:99--140`; materialization rejects an address-only chunk before attention in `src/pra_torch/core.py:232--270`. Selecting a chunk ID at L_R never licenses reuse of L_R ’s tensor at L_C : the identity is resolved to the destination layer’s independently projected K/V.

The complete geometry is concentrated in `src/pra_hf/native_geometry.py:59--374`. Lines 59–107 define four materialization modes and the three named research profiles. Lines 130–169 make raw and unique interval accounting, source lengths, and query offset part of the immutable plan. Lines 189–238 clamp expansion to one record; lines 310–374 normalize overlap before slicing each consumer layer’s own K/V.

The high-level path is in `src/pra_hf/model.py:932--1191`. `plan_native_materialization` separates routing anchors from selected chunk, asymmetric window, selected-record, or full-scope geometry. `native_next_token_logits` is the E0 diagnostic. `generate_with_native_plan` is the public fixed-plan path, and `debug_decode_with_native_plan` spells out prefill, local cache handoff, and one-token decode calls. All three use `src/pra_torch/hf/injection.py:494--529` to start the direct query after the longest active source extent.

At attention time, `src/pra_torch/hf/qwen.py:340--352` prepends selected native K/V to local K/V and records compact request-lifetime state before the family-native eager kernel. Llama and Gemma 3 inherit this shared path while overriding only family symbols and native scope. The trace owner is defined in `src/pra_torch/hf/adapter_base.py:107--129` and aggregated at `src/pra_torch/hf/injection.py:228--241`; no full K/V or attention tensor is retained for telemetry.

Portable streaming begins in `src/pra_hf/deployment.py:227--341`. The worker owns logical references, the cancellation criterion, and cleanup. Gateway mediation and SSE framing are separate at `src/pra_hf/gateway.py:131--231`, so a transport disconnect closes the iterator without exposing backing content. Tenant-scoped cache identity and local eviction are in `src/pra_hf/runtime.py:`

568--672. Cache identity includes source revision, positional and materialization signatures, and scope; equality is the reuse predicate rather than a heuristic URI match. Engine capability fields separately expose interval materialization, request lifetime, residency, tenant isolation, phase selection, and scheduler hints. An independent implementation should fail a required unsupported field before reference registration.

A.2 Gateway session implementation map

The independent capability axes begin at `src/prahf/deployment.py:38`: the capability record carries engine type, PRA level, prefix-cache mode, session state, incremental-message support, resource deltas, affinity, and explicit handles. Typed request serialization starts at line 129, and the profile-parameterized OpenAI adapter at line 242. The versioned defaults and validation are in `src/prahf/engine_profiles.py:1--90`; adding an engine requires an enum member and one conservative registry row before any feature override.

Ephemeral state is isolated in `src/prahf/gateway_session.py:186--434`. `resolve_turn` compares explicit full or delta history, derives resource operations, and computes the affinity key. `commit` advances fingerprints only after a successful decode. It stores resource IDs, digests, and metadata, never content tensors. The gateway execution order is visible at `src/prahf/gateway.py:147--457`: resolve and invalidate, prepare once, transform G00-G11, execute, commit trace state, and close explicitly. HTTP capability, session-debug, and session-delete handling follows at lines 460-538. The same registry entry points are surfaced by `src/prahf/runtime.py:1194--1220`, preventing the HTTP process from becoming the sole owner of session semantics.

A.3 Execution and deployment code map

Line references identify the artifact version accompanying this paper. `src/prahf_torch/execution.py:61--107` defines the four policy axes and their validation; lines 358-425 define tensor-free identities and row-local plans; lines 428-504 own request epochs, churn, and trace summaries; and lines 507-632 separate selection reuse from opaque materialization lifetimes. The important invariant is the type boundary: a selection controller returns URI/chunk spans, while only an engine resolver may return tensors or block handles.

The canonical typed-result admission policy is implemented at three layers. `src/prahf/adaptive_context_runtime.py:82--236` defines global and per-record token/byte limits, inheritance, deferral, and serialization. `src/prahf/progressive_context.py:58--66` defines the four lifecycle states; lines 347-391 define the public audit and lazy-region records; and lines 602-773 perform idempotent size-gated full indexing and selected-region native encoding. The facade in `src/prahf/runtime.py:1119--1268` applies automatic preparation at ingestion and exposes preparation, lazy promotion, and lifecycle inspection. This order is important: policy is resolved before model-native construction, and a gate rejection preserves the backing identity and recovery interfaces.

`src/prahf/hf_execution.py:49--174` is the family-independent HF bridge. At the routing layer it turns the current hidden representation into a logical plan. For shared scope it remaps the source-layer hits through `map_chunk_identities_to_layers`; each destination therefore receives its own native grouped-query K/V. The public request lock and precedence entry point are in `src/prahf/model.py:710--895`. Qwen, Llama, and Gemma call the same bridge from their thin attention adapter.

For a distributed implementation, begin with `src/prahf/deployment.py:100--174`: reproduce that JSON request without adding tensor fields. At lines 175-182, expose capabilities, session setup, generation, streaming, and close as separate methods. Then reproduce deterministic mediation in `src/prahf/gateway.py:24--178`: negotiate first, transform according to G00-G11 second, and append component-local trace events last. The HTTP process starts at line 180; the CLI binding is `src/prahf/gateway_cli.py:35--59`. This order keeps an unsupported native request from being mistaken for

successful text fallback. The agent bridge implementation is `src/pra_hf/agent_plugins.py:1--253`; its family-specific parsers terminate in the common logical request builder rather than engine calls. Frozen interval expansion and layer-native slicing are in `src/pra_hf/native_geometry.py:1--374`; the SDK facade delegates through `src/pra_hf/runtime.py` so users do not access adapter internals.

B Artifact Map

Artifact	Role
<code>src/pra_hf/runtime.py</code>	Unified facade, plans, materializer, cache, profiler, HF adapter, thin vLLM contract
<code>src/pra_torch/execution.py</code>	Engine-neutral policy axes, logical plans, precedence, lifecycle, and analytical routing count
<code>src/pra_hf/hf_execution.py</code>	Request-owned token/shared and token/per-layer HF execution bridge
<code>src/pra_hf/deployment.py, src/pra_hf/gateway.py</code>	JSON wire types, engine capabilities, E0/E1 adapters, G00–G11 mediation, and HTTP process
<code>src/pra_hf/engine_profiles.py, src/pra_hf/model_profiles/engine_profile_registry.json</code>	Versioned engine/cache/session defaults with conservative capability resolution
<code>src/pra_hf/gateway_session.py</code>	Tenant/session/model-isolated ephemeral engine handles, canonical and serialized prefix state, resource versions, history/resource deltas, affinity, and invalidation
<code>experiments/paper4_5_runtime/run_gateway_session_profile.py</code>	Five-turn prefix/session contract profile, two-cache architecture, traces, tables, and figures
<code>docs/papers/shared/results/paper4_5_runtime/prefix_preservation_results.csv</code>	Canonical per-turn logical-prefix and transport measurements; physical hits remain unknown
<code>src/pra_hf/agent_plugins.py</code>	DeepSeek Harness and Pi typed-event bridges
<code>src/pra_hf/native_geometry.py</code>	Frozen record-bounded intervals and layer-native K/V plans
<code>experiments/paper4_5_runtime/run_execution_policy_profile.py</code>	Five-seed multi-axis mechanism profile and trade-off figure
<code>experiments/paper4_5_runtime/run_cross_model_validation.py</code>	Restartable Qwen/Llama/Gemma semantic gates, physical accounting, reuse and multi-tenant controls, and cross-model figure
<code>experiments/paper4_5_runtime/run_layer_profile_calibration.py</code>	Corrected cross-model contiguous/sparse profile sweep, storage accounting, registry selection, and quality–cost figure
<code>src/pra_hf/layer_profiles.py</code>	Four-role layer schema, normalized selection, registry precedence, lifecycle, and missing-detail policy
<code>src/pra_hf/profile_benchmarks.py</code>	Versioned model/workload profile registry, normalized metrics, missingness status, and resolution trace
<code>src/pra_hf/model_profiles/pra_profile_benchmarks.json</code>	Packaged semantic and physical profile registry
<code>experiments/paper4_5_runtime/build_profile_matrix.py</code>	Deterministic registry and manuscript-table generator
<code>src/pra_hf/runtime_benchmark.py</code>	Structured eager/compiled and hierarchy microbenchmark
<code>src/pra_hf/agent_resources.py</code>	Typed resource identities and discovery
<code>src/pra_hf/agent_execution.py</code>	Schema validation and independent host authorization
<code>src/pra_hf/agent.py</code>	Task-aware turn orchestration and persistent agent facade
<code>src/pra_hf/session_service.py</code>	In-memory and atomic local logical session persistence
<code>src/pra_hf/task_context.py, src/pra_hf/task_scope.py</code>	Versioned task DAG, provenance, scope admission, and working-set lifecycle

Artifact	Role
<code>src/prahf/toolsets.py</code> , <code>src/prahf/tui.py</code> <code>src/prahf/capability_sdk.py</code>	Reusable tools and the resumable coding-agent terminal
<code>src/prahf/typed_context.py</code>	Callable and skill-folder ingestion, bounded palettes, and exact capability activation
<code>src/prahf/large_record_index.py</code>	Type-aware compact views and retrieval addresses with explicit token/ratio compression budgets
<code>src/prahf/adaptive_context_runtime.py</code> <code>src/prahf/progressive_context.py</code>	Independent typed, BM25, and embedding indexes, bounded fusion, exact selectors, and lifecycle accounting
<code>src/prahf/external_memory.py</code> <code>prahf-demo/prahf_runtime_productization.ipynb</code>	Scoped backing, serializable token/byte and per-type ingestion policy, materialization, transport accounting, and cursors
	Stable record views, native-index lifecycle audits, size-gated full backing, and lazy selected-region native encoding
	Resolver-neutral cold/warm/hot lifecycle and scoped caches
	Executed user workflow

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